

From Clicks to Footprints: Point-Supervised Building Segmentation and the Limits of Self-Training — A Pilot Study

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Abstract—Dense pixel-level annotation is the dominant cost in the preparation of training data for remote sensing semantic segmentation, and this cost scales directly with image resolution and the number of objects present in a scene. In this study, a weakly-supervised alternative is examined, in which a segmentation model is trained using only a small number of point clicks per image rather than a fully painted mask. A Partial Focal Cross-Entropy objective is formulated, under which gradients are propagated exclusively through annotated coordinates, and a rigorous experimental protocol is designed so that annotation budgets, random seeds, and evaluation splits are controlled throughout. It is found that ten clicks per tile, which corresponds to approximately 0.0004% of the pixels in a tile, recover close to seventy percent of the performance obtained under full supervision. It is further shown that a naive label-propagation strategy, in which clicks are grown into pseudo-masks before any training takes place, is outperformed by the trained point-supervised model itself, across ten separate propagation configurations. A confidence-gated self-training procedure is then applied, in which a committee of point-supervised teachers is used to generate pseudo-labels for a student network; a first round of self-training is found to improve foreground Intersection-over-Union by 0.017, whereas a second round yields a smaller gain of 0.005, and the underlying cause of this saturation is identified through a precision analysis of the confidence gate. The reported results are averaged, where applicable, over three simulated annotators, and the test set is evaluated only once, at the conclusion of the study, so that the reported figures reflect genuine generalisation rather than repeated tuning against the same held-out data.

Index Terms—weakly-supervised semantic segmentation, point supervision, remote sensing, building footprint extraction, focal loss, self-training, pseudo-labelling, superpixels

I. INTRODUCTION

Semantic segmentation of remote sensing imagery is used widely in urban planning, disaster response, and infrastructure monitoring, and among its most common applications is the automatic extraction of building footprints from aerial or satellite photography. Convolutional architectures such as U-Net, when trained with fully-supervised dense masks, are capable of producing accurate building outlines. However, the creation of a dense mask requires that every pixel in every training image be labelled by hand, and this requirement is widely regarded as the primary bottleneck in the deployment of segmentation models to new regions or new sensor types, since the cost of annotation is not offset by any corresponding gain in accuracy once a sufficient number of pixels has been labelled.

A natural question is therefore raised: how much of the performance of a fully-supervised model can be recovered if only a handful of point annotations are provided per image, rather than a complete mask? This question is not new, and it has been examined previously in the general computer vision literature under the heading of point supervision [1]. Its application to remote sensing imagery, however, raises considerations that are specific to the domain, since buildings in aerial photography are frequently small, tightly packed, and visually similar in colour and texture to paved surfaces such as roads and car parks. It is not obvious, prior to experimentation, whether a model trained on a handful of clicks per tile will be able to separate these two classes reliably, or whether classical unsupervised techniques, such as superpixel segmentation, can be used to expand a small number of clicks into a useful pseudo-mask before any learning takes place.

This study is motivated by these considerations, and it reports on a pilot investigation that was carried out on the Massachusetts Buildings Dataset [2], using a U-Net architecture with a ResNet-34 encoder. The central research question is stated plainly: given ten point clicks per training tile, how close can a segmentation model come to the accuracy of a model that is trained with full pixel-level supervision, and can this gap be narrowed further through label propagation or self-training, without any additional human annotation effort?

The contributions of this study are summarised as follows.

- A Partial Focal Cross-Entropy loss function is formulated and implemented, under which the gradient is constrained exclusively to annotated point coordinates, so that unlabelled pixels do not contribute noise to the optimisation process.
- A controlled experimental protocol is designed, in which point annotations are sampled once and frozen as annotation manifests, in which the weakly-supervised training pipeline is constructed so that it is physically unable to access dense ground truth, and in which three simulated annotators are used so that the sensitivity of the results to the particular pixels that are clicked can be measured directly.
- Three families of label-propagation methods, namely Gaussian heatmap splatting, SLIC superpixel propagation, and feature-space k-nearest-neighbour propagation, are evaluated across a sweep of annotation budgets, and

it is shown that none of them, in this pilot study, is able to exceed the accuracy of the trained point-supervised model itself.

- A confidence-gated self-training procedure is implemented and evaluated across two rounds, and the observed saturation of performance after the first round is explained through an analysis of the calibration of the teacher committee relative to the self-trained student.

The remainder of this paper is organised as follows. Section II reviews related work in weakly-supervised segmentation, point supervision, and self-training. Section III describes the dataset and formalises the problem. Section IV presents the methodology, including the point-sampling protocol, the loss function, the model architecture, the label-propagation strategies, and the self-training procedure. Section V describes the experimental protocol and the safeguards that were put in place to ensure that the reported results are trustworthy. Section VI reports the results. Section VII discusses the findings and offers an interpretation of why label propagation was found to under-perform and why self-training was found to saturate. Section VIII discusses the limitations of the study and directions for future work, and Section IX concludes the paper.

II. RELATED WORK

A. Weakly-Supervised Semantic Segmentation

Weakly-supervised semantic segmentation seeks to reduce the annotation burden associated with dense pixel labelling by substituting cheaper forms of supervision, such as image-level tags, bounding boxes, scribbles, or points. Among these, point supervision was formalised by Bearman *et al.* [1], who demonstrated that a small number of clicks per object, combined with an objectness prior, could substantially narrow the gap between weak and full supervision on natural images. The present study follows a similar spirit, but restricts itself to raw point clicks without an external objectness prior, so that the contribution of the loss function and of the propagation strategy can be isolated more cleanly.

B. Superpixel and Feature-Based Label Propagation

Superpixel algorithms, of which SLIC [6] is among the most widely used, group visually homogeneous and spatially contiguous pixels into small regions, and have been employed in prior work to expand sparse annotations into denser pseudo-masks under the assumption that object boundaries coincide with superpixel boundaries. Feature-space propagation methods extend this idea by measuring similarity in a learned or pretrained embedding space, rather than in raw colour space, and propagating labels to the nearest neighbours of an annotated pixel. Both families of method are examined in this study as candidate pre-training pseudo-label generators.

C. Focal Loss

The focal loss, introduced by Lin *et al.* [5] in the context of dense object detection, was designed to address class imbalance by down-weighting the contribution of well-classified

examples and concentrating gradient signal on hard or misclassified ones. In this study, focal loss is combined with a partial, coordinate-gated cross-entropy term so that it operates only over the sparse set of annotated points, and its interaction with annotation density is examined directly.

D. Self-Training and Pseudo-Labeling

Self-training, in which a model trained on labelled data is used to generate pseudo-labels for unlabelled data, and is then retrained on the union of the two, has a long history in semi-supervised learning [7], and has been extended through consistency regularisation in methods such as Mean Teacher [8] and FixMatch [9]. The present study applies a confidence-gated self-training procedure in which a committee of independently seeded point-supervised teachers is used to filter pseudo-labels before a student network is trained, and the calibration of the resulting pseudo-labels is examined explicitly, which is a step that is not always reported in the self-training literature.

E. Building Footprint Extraction

The U-Net architecture [3], frequently paired with a ResNet encoder [4] pretrained on natural images, is a standard choice for building footprint extraction from aerial imagery, and it is adopted without modification in this study, so that the reported findings can be attributed to the supervision strategy rather than to architectural novelty.

III. DATASET AND PROBLEM FORMULATION

A. Massachusetts Buildings Dataset

All experiments in this study are conducted on the Massachusetts Buildings Dataset [2], which consists of aerial imagery of the state of Massachusetts, captured at a spatial resolution of approximately one metre per pixel, together with corresponding binary building masks. The dataset is partitioned into 137 training tiles, 4 validation tiles, and 10 test tiles, each of size 1500×1500 pixels. For training and validation purposes, tiles are divided into 256×256 patches, whereas final test-set evaluation is carried out at the full tile resolution through a windowed inference procedure, so that the reported test metric reflects performance over an entire, realistically sized scene rather than over isolated crops.

B. Problem Formulation

Let $Y \in \{0, 1\}^{H \times W}$ denote the dense ground-truth mask of a tile, where a value of one indicates a building pixel and a value of zero indicates background. A sparse annotation mask $M \in \{-1, 0, 1\}^{H \times W}$ is constructed by sampling exactly N coordinates per class from Y , where a value of -1 indicates that a pixel is unannotated, and values of 0 and 1 indicate confirmed background and foreground clicks respectively. The learning objective is to recover a predictor $\hat{Y} = f_{\theta}(X)$, trained using supervision from M alone, that approximates Y as closely as possible when evaluated over the full, dense image space, rather than merely over the small set of coordinates on which it was trained. This distinction between training supervision and evaluation target is central to the present study,

and it motivates the two separate metrics, Point-IoU and Dense (full-tile) foreground IoU, that are used throughout.

IV. METHODOLOGY

A. Point Annotation Simulation

Point annotations are simulated rather than collected from human annotators, so that the annotation budget, denoted N , can be varied in a controlled manner. For a given tile and a given value of N , exactly N background pixels and N foreground pixels are sampled uniformly at random from the corresponding regions of the dense mask; where a class contains fewer than N pixels, sampling is performed with replacement, so that the constraint of exactly N points per class is always satisfied. Crucially, annotation sets are sampled once per tile and are then frozen and exported as JSON manifests, in the same form in which a human annotator's clicks would be recorded, and these manifests are reused across all subsequent experiments rather than being re-sampled at every training run. Annotation budgets are additionally constructed to be nested, such that the set of clicks used at $N = 5$ is a subset of the set used at $N = 10$, which models the incremental effort of an annotator who is asked to provide additional clicks after an initial pass.

B. Partial Focal Cross-Entropy Loss

In order to constrain learning exclusively to annotated coordinates, a Partial Focal Cross-Entropy loss, denoted $\mathcal{L}_{\text{ptCE}}$, is formulated. Let p_t denote the model's predicted probability for the true class at a given coordinate, and let γ denote the focal modulation parameter. The loss is defined as

$$\mathcal{L}_{\text{ptCE}}(\theta) = \frac{\sum_{i,j} \mathbb{I}(M_{i,j} \neq -1) \cdot (1 - p_t)^\gamma \log(p_t)}{\sum_{i,j} \mathbb{I}(M_{i,j} \neq -1) + \epsilon} \quad (1)$$

where $\mathbb{I}(\cdot)$ is a spatial gating indicator that is equal to one at annotated coordinates and zero elsewhere, and ϵ is a small constant that prevents division by zero when a mask contains no annotated points. Setting $\gamma = 0$ reduces the objective to a standard Partial Cross-Entropy loss, which is used as a reference configuration throughout this study, alongside the focal-modulated variant with $\gamma = 2$.

C. Model Architecture

A U-Net segmentation architecture [3], with a ResNet-34 encoder [4] initialised from ImageNet pretraining, is used in all experiments. Input images are normalised using ImageNet channel statistics prior to being passed to the encoder, and the same architecture and initialisation procedure are used across the fully-supervised, point-supervised, and self-trained configurations, so that any difference in performance between configurations can be attributed to the supervision strategy rather than to architectural variation.

D. Label Propagation Strategies

Three families of label-propagation method are implemented, under the hypothesis that a small number of clicks might be expanded into a denser pseudo-mask prior to training, and thereby provide the network with a stronger training signal than raw, isolated clicks.

- 1) **Gaussian heatmap splatting.** A two-dimensional Gaussian kernel of fixed radius σ is centred at each clicked coordinate, and pixels that fall within this radius are assigned the label of the corresponding click.
- 2) **SLIC superpixel propagation.** The SLIC algorithm [6] is applied to each input tile in order to obtain a partition of the image into spatially contiguous, visually homogeneous superpixels. Where a clicked coordinate falls within a given superpixel, the corresponding label is propagated to every pixel within that superpixel. Superpixels that receive clicks from both classes are left unlabelled, since the presence of conflicting clicks indicates that the corresponding region straddles an object boundary and cannot be resolved unambiguously.
- 3) **Feature-space k-nearest-neighbour propagation.** Pixel-wise feature representations are extracted using a frozen, pretrained convolutional encoder, and each unlabelled pixel is assigned the label of its nearest annotated neighbour in this feature space.

In addition to these three individual methods, a consensus pseudo-mask is constructed by retaining, at each pixel, the majority label across the three methods, and discarding pixels for which no majority is reached.

E. Self-Training Protocol

A confidence-gated self-training procedure is designed, in which three point-supervised teacher networks, each trained independently using a different random seed, are combined into a teacher committee. For every unlabelled pixel in the training set, the committee's mean predicted probability is computed, and a pseudo-label is retained only where this mean probability exceeds a fixed confidence threshold, so that only high-confidence predictions are used to supervise the subsequent training round. A student network is then trained using a combination of the original point clicks and the confidence-gated pseudo-labels. This procedure is repeated for a second round, in which the student from the first round is added to the teacher committee, so that the committee for round two consists of the three original teachers together with the round-one student.

V. EXPERIMENTAL PROTOCOL

In order to ensure that the reported results are trustworthy and are not the product of data leakage, favourable random seeding, or repeated tuning against the test set, the following safeguards are adopted throughout this study.

- **Frozen, seeded annotation.** Point clicks are sampled once per tile and are exported as fixed JSON manifests, which are reused identically across every experiment; clicks are never re-sampled between runs.

- **Leak-proof construction.** The weakly-supervised training pipeline is constructed such that it never has access to the dense ground-truth mask during training; dense masks are used exclusively for evaluation, so that leakage of dense supervision into the training signal is not merely avoided through discipline, but is rendered structurally impossible.
- **Annotation-budget accounting.** Click sets at different values of N are constructed to be nested, modelling the effort of an annotator who incrementally adds further clicks rather than re-annotating from scratch.
- **Full-tile, dataset-level evaluation.** Foreground IoU is computed over entire 1500×1500 test tiles using windowed inference, rather than over isolated 256×256 patches, and the background-inflated mean IoU that conflates foreground and background performance is avoided in favour of foreground-specific IoU.
- **Three simulated annotators.** Each point-supervision experiment is repeated with three independently seeded click sets, and results are reported as the mean and standard deviation across these three seeds, so that the reported spread reflects genuine annotator variance rather than training noise alone.
- **Single test-set evaluation.** The test set is evaluated exactly once, at the conclusion of the study, for the purpose of producing the final results table; all intermediate development and tuning decisions are made using the validation split alone.
- **Cross-session reproducibility.** All random seeds are fixed in advance, and the complete set of results was independently re-verified through a fresh execution of the training pipeline, in order to confirm that the reported figures are stable and are not an artefact of a single fortunate run.

VI. RESULTS

A. Point Supervision Recovers Most of Full Supervision

The central finding of this study is that a small number of point clicks per tile is sufficient to recover a substantial fraction of fully-supervised performance. As shown in Fig. 1, an early checkpoint of the point-supervised model, trained using ten clicks per tile, achieves a validation foreground IoU of 0.453 ± 0.007 , compared with 0.657 ± 0.004 for the fully-supervised model trained on the complete dense mask. At the conclusion of training, and when evaluated on the held-out test set, the point-supervised baseline achieves a foreground IoU of 0.494 ± 0.002 , against a fully-supervised ceiling of 0.711 ± 0.001 , which corresponds to the recovery of approximately seventy percent of fully-supervised performance from an annotation budget that is smaller by several orders of magnitude, at approximately 0.0004% of the pixels in each tile. The spread across three simulated annotators is narrow, at ± 0.002 on the test set, which indicates that the result is largely insensitive to the particular pixels that happen to be clicked, rather than being an artefact of a fortunate choice of annotation coordinates.

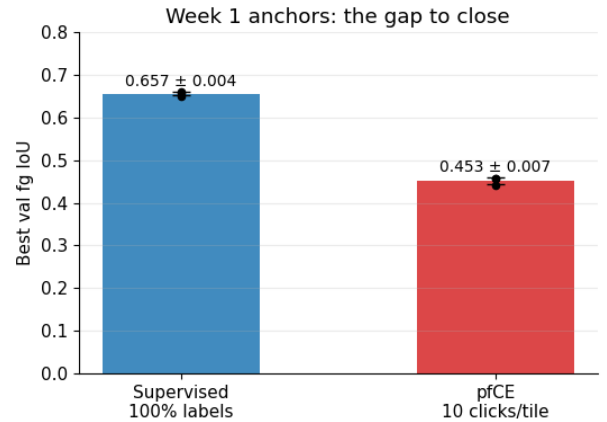


Fig. 1: Validation foreground IoU for the fully-supervised model (100% of pixels labelled) compared against the point-supervised Partial Focal Cross-Entropy model (10 clicks per tile), reported as mean $\pm$ standard deviation over three simulated annotators.

B. The Click-Memorisation Trap

Although the point-supervised model narrows the gap to full supervision substantially, its training dynamics reveal a systematic weakness. Fig. 2 presents the validation foreground IoU of both configurations as a function of training epoch. The fully-supervised curve rises smoothly and plateaus at a stable value. The point-supervised curve, by contrast, rises during the early epochs but reaches a peak at approximately epoch twenty-five, after which it declines gradually, even though the training loss continues to fall throughout. This pattern indicates that the model, having exhausted the information that can usefully be extracted from its small set of clicked pixels, begins to memorise the exact identity of those pixels rather than to generalise the appearance of buildings across the image, a phenomenon that is referred to in this study as click memorisation. This finding underlines the importance of early stopping, or of an equivalent regularisation strategy, whenever point supervision is used at a low annotation budget.

C. Label Propagation Under-performs the Trained Model

A central hypothesis examined in this study is that a pseudo-mask, grown from a small number of clicks prior to training, might provide a stronger training signal than the raw clicks themselves. This hypothesis was tested through ten separate configurations, spanning Gaussian heatmap splatting, SLIC superpixel propagation, and feature-space k-nearest-neighbour propagation, each evaluated across a sweep of annotation budgets from five to one hundred clicks per class per tile. As shown in Fig. 3, every propagation method examined remains below the dashed reference line, which denotes the foreground IoU of 0.494 achieved by the trained, point-supervised model. The strongest propagation method, feature-space k-nearest-neighbour, reaches a pseudo-label foreground IoU of only 0.234 even at the largest annotation budget tested, which corresponds to a labelled coverage of approximately 79% of

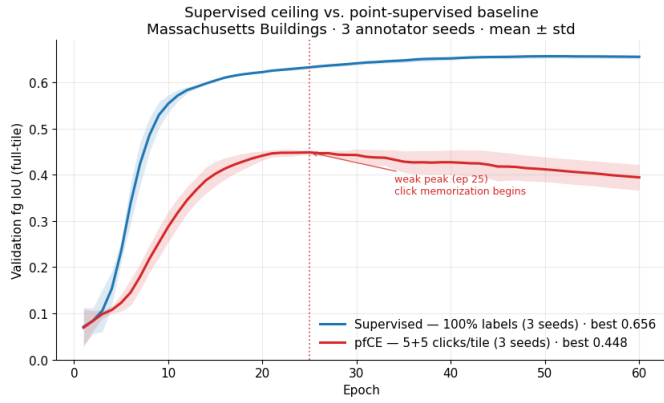


Fig. 2: Validation foreground IoU across training epochs for the fully-supervised model and the point-supervised (5+5 clicks per tile) model, each averaged over three annotator seeds, with shaded regions indicating one standard deviation. The point-supervised curve peaks near epoch 25 and subsequently declines, a pattern attributed to click memorisation.

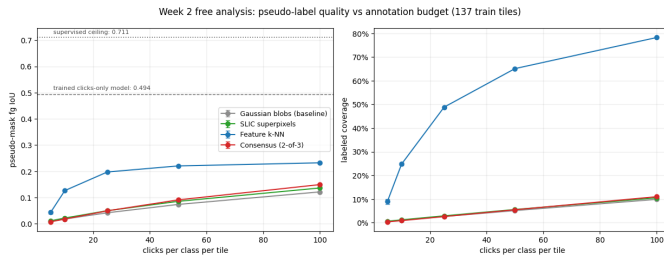


Fig. 3: Left: pseudo-mask foreground IoU as a function of annotation budget, for each of the three propagation methods and their consensus, together with reference lines denoting the fully-supervised ceiling and the trained point-supervised model. Right: labelled coverage achieved by each method at the same annotation budgets.

the tile, whereas SLIC superpixel propagation and Gaussian splatting remain considerably lower, with pseudo-label IoU below 0.15 across the entire sweep, at a labelled coverage of approximately 10% of the tile.

Fig. 4 presents qualitative examples of the pseudo-masks produced by each method at a budget of twenty-five clicks per class per tile. The Gaussian and SLIC pseudo-masks are visibly sparse and fragmented, and the feature-space k-nearest-neighbour method, while covering a substantially larger area of each tile, is observed to expand into regions of asphalt and shadow that share a similar grey appearance with rooftops, which is consistent with the low precision implied by its correspondingly modest IoU.

D. Self-Training Improves Performance for One Round

Table I reports the principal results of this study on the held-out test set. Confidence-gated self-training, in which a committee of three point-supervised teachers generates pseudo-labels at a precision of 82.7% under the chosen confidence threshold,

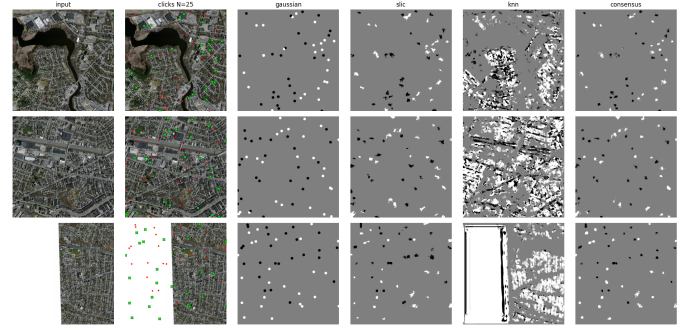


Fig. 4: Qualitative comparison of the three propagation methods and their 2-of-3 consensus, at an annotation budget of 25 clicks per class per tile. White denotes a foreground pseudo-label, black a background pseudo-label, and grey an unlabelled pixel.

TABLE I: Main Results on the Held-Out Test Set

Method	Supervision	Fg. IoU	F1
Supervised ceiling	100% of pixels	0.711 ± 0.001	0.831
Self-training R2 (committee)	30 clicks/tile + pseudo	0.516	0.681
Self-training R1	30 clicks/tile + pseudo	0.511	0.676
Partial Focal CE (baseline)	10 clicks/tile	0.494 ± 0.002	0.662

is found to improve the foreground IoU of the student network from 0.494 to 0.511 after a single round, an absolute gain of 0.017. A second round of self-training, in which the round-one student is incorporated into the teacher committee, yields a further but smaller gain, raising the foreground IoU to 0.516, an increment of only 0.005 relative to the first round. Fig. 5 presents qualitative examples on three test tiles, comparing the fully-supervised ceiling, the ten-click baseline, and the round-two self-trained student; the self-trained model is observed to recover a number of buildings that are missed by the raw point-supervised baseline, while remaining visibly less complete than the fully-supervised reference.

VII. DISCUSSION

A. Why Does Label Propagation Fail on This Dataset?

The finding that label propagation under-performs the trained model, despite its intuitive appeal, is attributed to the spatial resolution and visual characteristics of the dataset under study. At a ground sampling distance of approximately one metre per pixel, individual buildings occupy regions of the order of twenty by twenty pixels, and grey rooftops are frequently similar in colour and texture to paved roads and car parks in the same neighbourhood. Superpixel and colour-based propagation methods, which rely on low-level visual homogeneity, are consequently unable to distinguish a rooftop superpixel from a road superpixel in a large number of cases. Feature-space propagation, which relies on a frozen, pretrained encoder, fares somewhat better, since generic convolutional features carry some information beyond raw colour, but it remains substantially below the trained model, because the encoder has not been trained specifically to separate these

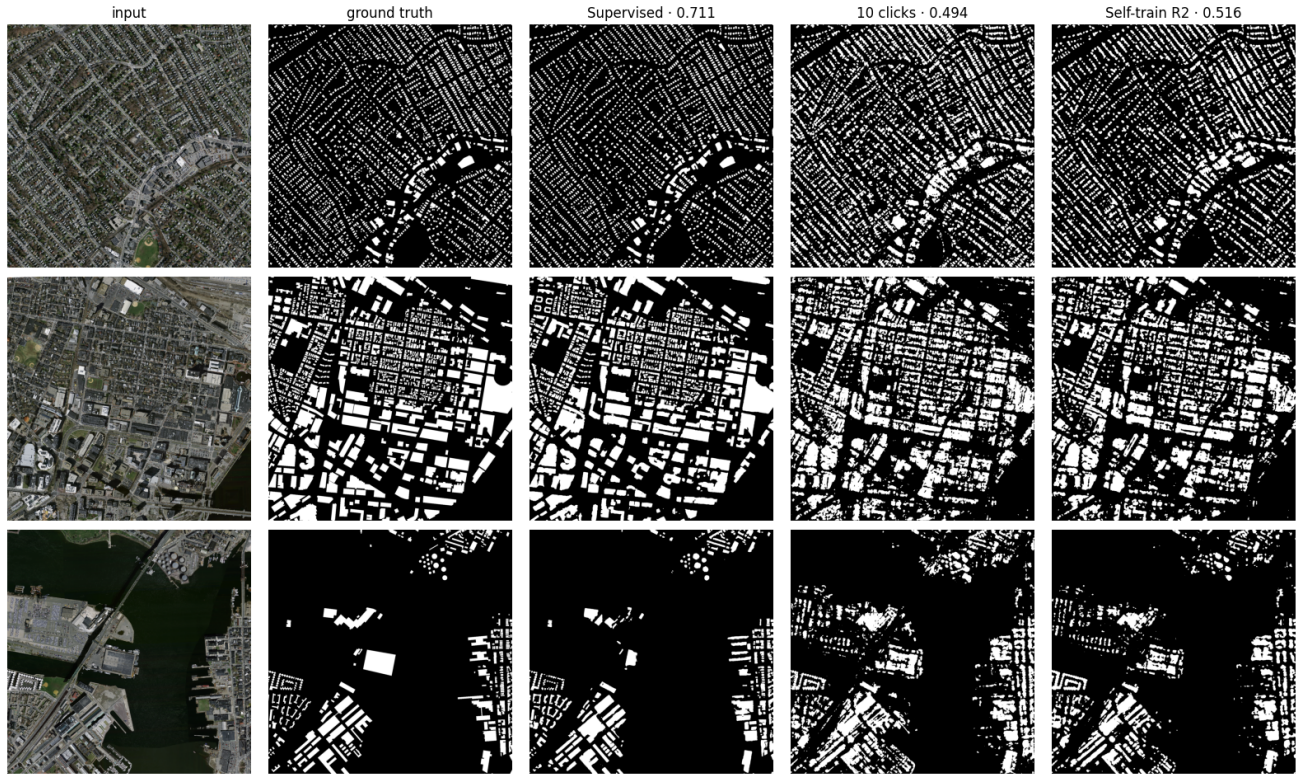


Fig. 5: Qualitative comparison on three test tiles: input imagery, dense ground truth, the fully-supervised prediction (foreground IoU 0.711), the ten-clicks-per-tile point-supervised prediction (0.494), and the round-two self-trained prediction (0.516).

two classes; it is only through end-to-end training, in which gradients are permitted to reshape the feature space toward the specific discrimination that is required, that the buildings-versus-pavement distinction is learned effectively. This observation suggests that, for datasets of comparable resolution and visual ambiguity, propagation-before-training strategies of the kind evaluated in this study are unlikely to substitute for direct training on the raw clicks, and that any benefit of such strategies is more likely to be realised when they are applied after an initial round of training, once the network’s features have already been adapted to the target classes.

B. Why Does Self-Training Saturate After One Round?

The saturation of the self-training procedure after a single round is explained through a precision analysis of the confidence gate. The original teacher committee, trained using only sparse point clicks and focal-modulated loss, is found to be comparatively well calibrated, in the sense that its high-confidence predictions are correct 82.7% of the time. The round-one student, however, having been trained on a combination of clicks and pseudo-labels, is found to become considerably more confident in its own predictions without a corresponding improvement in correctness, and the precision of its high-confidence predictions, measured at the identical threshold, falls to 58.8%. This pattern is consistent with the student inheriting and reinforcing the specific errors present in its teachers’ pseudo-labels, a phenomenon that is commonly

described as confirmation bias in the self-training literature. When the round-one student is incorporated into a new committee alongside the three original teachers, the precision of the resulting confidence gate is restored to 80.4%, which is close to the original teacher precision, and this restoration of calibration is offered as the explanation for why round two produces a further, though smaller, improvement: the committee is able to partially correct for the overconfidence of any single member, but the overall information available to the pipeline is not substantially increased, since no new source of supervision, beyond the original clicks, has been introduced. This suggests that the observed gain of self-training in this setting is bounded not primarily by the capacity of the student network, but by the amount of genuinely new information that can be extracted from a fixed committee of teachers trained on a fixed, small set of point annotations.

C. Practical Implications

Taken together, these findings suggest a specific and practical recommendation for practitioners who face the cost of dense annotation in remote sensing segmentation: a small, carefully budgeted set of point clicks, combined with a partial, focal-modulated loss and a single round of confidence-gated self-training, is capable of recovering a large fraction of fully-supervised accuracy, at an annotation cost that is smaller by several orders of magnitude. Attempting to close the remaining gap through additional rounds of self-training, or through pre-

training label propagation, is not supported by the evidence collected in this study, and effort may be better directed toward alternative strategies, such as the incorporation of a small number of additional, actively selected annotations, or the use of consistency-regularisation techniques that do not rely on hard pseudo-labels.

VIII. LIMITATIONS AND FUTURE WORK

Several limitations of this pilot study are acknowledged. First, all experiments were conducted on a single dataset, the Massachusetts Buildings Dataset, and it remains to be established whether the findings generalise to remote sensing imagery of different resolution, geography, or building density. Second, a single model architecture, namely U-Net with a ResNet-34 encoder, was used throughout, and the extent to which the reported findings are specific to this architecture, as opposed to being a general property of point-supervised segmentation, has not been established. Third, while the point-supervised baseline was evaluated across three annotator seeds, the self-training rounds were each evaluated using a single seed, owing to the computational cost of the multi-round pipeline, and the variance of the self-training results across annotator seeds is accordingly not yet known. Finally, a gap of approximately 0.20 in foreground IoU remains between the best self-trained configuration and the fully-supervised ceiling, and this gap is identified as the principal target for future work.

Building on these limitations, several directions for future work are proposed. Consistency-regularisation techniques, such as those used in Mean Teacher [8] and FixMatch [9], which enforce agreement between a model’s predictions under different perturbations of the same input rather than relying on discrete pseudo-labels, may be less susceptible to the confirmation-bias effect identified in Section VII-B, and are proposed as a natural extension of the present self-training procedure. The use of stronger or task-adapted encoders, potentially pretrained on remote sensing imagery rather than natural images, may improve the quality of feature-space label propagation and is proposed as a further avenue of investigation. A systematic sweep of the annotation budget, from a single click per class per tile up to one hundred, is proposed in order to characterise the full annotation-efficiency curve, rather than the two budgets examined in the present study. Finally, the extension of the present protocol to additional datasets and to object classes beyond buildings is proposed as a necessary step before the present findings can be regarded as general properties of point-supervised remote sensing segmentation, rather than properties specific to the dataset examined here.

IX. CONCLUSION

This study examined the extent to which a building segmentation model can be trained using only a small number of point clicks per image, rather than a fully painted dense mask, on the Massachusetts Buildings Dataset. A Partial Focal Cross-Entropy loss was formulated in order to constrain learning to annotated coordinates, and a rigorous experimental protocol, involving frozen annotation manifests, leak-proof

pipeline construction, and repeated evaluation across simulated annotators, was designed in order to ensure that the reported findings are trustworthy. It was found that ten clicks per tile are sufficient to recover approximately seventy percent of fully-supervised performance, that the point-supervised model is prone to a memorisation effect that is not observed under full supervision, that ten separate label-propagation strategies fail to exceed the accuracy of the trained model itself, and that a confidence-gated self-training procedure yields a measurable but rapidly saturating improvement, the saturation of which was traced to a loss of calibration in the self-trained student relative to its teachers. These findings are offered as evidence that point supervision, combined with focal loss modulation and a single round of self-training, represents a practical and substantially cheaper alternative to full dense annotation for building footprint extraction, while also identifying the specific mechanisms, namely click memorisation and self-training confirmation bias, that must be addressed in order to narrow the remaining gap to full supervision.

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